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TEST BOOKLET NO-26004

PAPER -II

Time Allowed: 3:00 hrs

ELECTRICAL ENGINEERING

Maximum Marks: 300

INSTRUCTIONS TO CANDIDATES

Read the instructions carefully before answering the questions: -

1. This Test Booklet consists of 16 (sixteen) pages and has 75 (seventy five) items (questions).
2. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS BOOKLET **DOES NOT** HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET.
3. Please note that it is the candidate's responsibility to fill in the Roll Number and other required details carefully and without any omission or discrepancy at the appropriate places in the OMR Answer Sheet and the Separate Answer Booklet. Any omission/discrepancy will render the OMR Answer Sheet and the Separate Answer Booklet liable for rejection.
4. Do not write anything else on the OMR Answer Sheet except the required information. Before you proceed to mark in the OMR Answer Sheet, please ensure that you have filled in the required particulars as per given instructions.
5. Use **only Black/Blue Ball Point Pen** to fill the OMR Answer Sheet.
6. This Test Booklet is divided into 4 (four) parts – **Part – I, Part - II , Part – III** and Part IV
7. All four parts are **Compulsory**.
8. **Part-I consists of Multiple Choice-based Questions.** The answers to these questions have to be marked in the **OMR Answer Sheet** provided to you.
9. **Part - II ,Part – III and Part IV consists of Conventional Questions.** The answers to these questions have to be written in the **Separate Answer Booklet** provided to you.
10. In Part-I, each item (question) comprises of 04 (four) responses (answers). You are required to select the response which you want to mark on the OMR Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose **ONLY ONE** response for each item.
11. After you have completed filling in all your responses on the OMR Answer Sheet and the Answer Booklet(s) and the examination has concluded, you should hand over to the Invigilator **only the OMR Answer Sheet and the Answer Booklet(s)**. You are permitted to take the Test Booklet with you.
12. **Penalty for wrong answers in Multiple Choice-based Questions:**
THERE WILL BE **PENALTY** FOR WRONG ANSWERS MARKED BY A CANDIDATE.
 - (i) There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, **one-third** of the marks assigned to the question will be deducted as penalty.
 - (ii) If a candidate gives more than one answer, it will be treated as a **wrong answer** even if one of the given answers happens to be correct and there will be same penalty as above to the question.
 - (iii) If a question is left blank. i.e., no answer is given by the candidate, there will be **no penalty** for that question.

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PART – I

MULTIPLE CHOICE BASED QUESTIONS

Instructions for Questions 1 to 50:

- Choose the correct answer for the following questions
- Each question carries 3 marks

(50 x 3 = 150 marks)

1. The Voltage across 1.1 KW toaster that produces a current of 10 A is:
A) 11 KV
B) 1100 KV
C) 110 KV
D) 11 V
2. A tuned amplifier has a maximum output at 4 MHz with a quality factor of 50. The band with and half power frequencies are, respectively
A) 80 kHz and 4.04 MHz; 3.96 MHz
B) 80 kHz and 4.08 MHz; 3.92 MHz
C) 40 kHz and 4.04 MHz; 3.96 MHz
D) 40 kHz and 4.08 MHz; 3.92 MHz
3. A network has 12 branches and 8 independent loops. How many nodes are there in the network?
A) 5
B) 17
C) 19
D) 4
4. In a half-wave rectifier, if an AC supply is 60 Hz, then what is AC ripple frequency at output?
A) 30 Hz
B) 60 Hz
C) 120 Hz
D) 15 Hz
5. Select the statement which doesn't hold true:
A) A super-node consists of two non-reference nodes connected by a voltage source.
B) Nodal analysis is the application of KCL at the non-reference nodes.
C) A super-mesh consists of two meshes that have a voltage source in common.
D) Mesh analysis is the application of KVL around meshes in a planar circuit.

6. If an input periodic signal with non-zero dc component is impressed upon a high-pass RC circuit, what will be the dc component in the output waveform?
- A) Zero
 - B) It depends on the value of the capacitor
 - C) It depends on the value of resistor
 - D) Same as that in the input
7. A Wheatstone bridge has $R_1 = R_3 = 1 \text{ k}\Omega$. R_2 is adjusted until no current flows through the galvanometer. At that point, $R_2 = 3.2 \text{ k}\Omega$. The value of the unknown resistance is: network has 12 branches and 8 independent loops. How many nodes are there in the network?
- A) $2.5 \text{ k}\Omega$
 - B) $3.2 \text{ k}\Omega$
 - C) $5 \text{ k}\Omega$
 - D) $3 \text{ k}\Omega$
8. A uniform transmission line of characteristic impedance $100 \text{ }\Omega$ and feeding a purely resistive load of $500 \text{ }\Omega$ uses single stub matching. The stub is placed at a distance d from the load. The VSWR on the length d and on the stub itself will be
- A) 0 and 0 respectively
 - B) 1 and 1 respectively
 - C) 5 and 5 respectively
 - D) 5 and ∞ respectively
9. The output voltage of a 5-bit D/A binary ladder that has digital output of 11010 (Assuming $0 = 0 \text{ V}$ and $1 = +10 \text{ V}$) is
- A) 3.4375 V
 - B) 6.0 V
 - C) 8.125 V
 - D) 9.6875 V
10. The number of unused states in a 4-bit Johnson counter is:
- A) 2
 - B) 4
 - C) 8
 - D) 12

11. A bridge rectifier uses 9V AC input voltage, The diodes are ideal, what is the dc output voltage?
- A) 12.726 V
 - B) -12.726 V
 - C) 9 V
 - D) 8.1 V
12. The output of a moore sequential machine is a function of:
- A) All present states of the machine
 - B) All inputs
 - C) All combination of inputs and present state
 - D) Few combinations of inputs and present state
13. A current through a branch in a linear network is 2 A when the input source voltage is 10 V. If the voltage is reduced to 1 V and the polarity is reversed. The current through the branch is:
- A) -2 A
 - B) 2.5 A
 - C) 2 A
 - D) -0.2 A
14. In an RC coupled amplifier, the gain decreases in the frequency response due to the
- A) Coupling capacitor at low frequency and bypass capacitor at high frequency.
 - B) Coupling capacitor at high frequency and bypass capacitor at low frequency.
 - C) Coupling device capacitance at low frequency and coupling capacitor at high frequency.
 - D) Device junction capacitance at high frequency and coupling capacitor at low frequency.
15. How is inversion achieved using EX-OR gate?
- A) Giving input signal to the two input lines of the gate tied together
 - B) Giving input to one input line and logic zero to the other line.
 - C) Giving input to one input line and logic one to the other line.
 - D) Inversion can't be achieved using EX-OR gate.
16. Select the false statement in the following:
- A) In a difference amplifier the output is proportional to the difference of the two inputs.
 - B) In a non-inverting amplifier, the output voltage is a negative multiple of the input.
 - C) An ideal op-amp has an infinite input resistance, a zero-output resistance, and an infinite gain.
 - D) Instrumentation amplifier is an extension of the difference amplifier in that it amplifies the difference between its input signals.

17. A quantity that contains all the power information in the given load is:
- A) Complex Power
 - B) Power factor
 - C) Average Power
 - D) Reactive Power
18. The primary advantage of a crystal oscillator is that:
- A) It can oscillate at any frequency
 - B) It gives a high output voltage
 - C) Its frequency of oscillations remains almost constant
 - D) It operates on a very low dc supply voltage
19. The Thevenin impedance of a network seen from the load terminals is $50 + j20 \Omega$. For maximum power transfer, the load impedance must be:
- A) $-50 - j20 \Omega$
 - B) $-50 + j20 \Omega$
 - C) $50 - j20 \Omega$
 - D) $50 + j20 \Omega$
20. The intrinsic impedance of copper at high frequencies is:
- A) Purely resistive
 - B) Purely inductive
 - C) Complex with a capacitive component
 - D) Complex with an inductive component
21. A Y-connected balanced three-phase generator with an impedance of $0.4 + j0.3 \Omega$ per phase is connected to a Y-connected balanced load with an impedance of $24 + j19 \Omega$ per phase. The line joining the generator and the load has an impedance of $0.6 + j0.7 \Omega$ per phase. If the positive sequence of the source voltages is considered with $V_{an} = 120 \angle 30^\circ \text{ V}$, then the line voltages are:
- A) $120 \angle 60^\circ \text{ V}, 120 \angle 120^\circ \text{ V}, 120 \angle 180^\circ \text{ V}$
 - B) $207.8 \angle 60^\circ \text{ V}, 207.8 \angle -60^\circ \text{ V}, 207.8 \angle -180^\circ \text{ V}$
 - C) $120 \angle 60^\circ \text{ V}, 120 \angle -60^\circ \text{ V}, 120 \angle -180^\circ \text{ V}$
 - D) $207.8 \angle 60^\circ \text{ V}, 207.8 \angle 120^\circ \text{ V}, 207.8 \angle 180^\circ \text{ V}$
22. In a $\Delta - \Delta$ system, a phase voltage of 110 V produces a line voltage of:
- A) 55 V
 - B) 110 V
 - C) 41 V
 - D) 173 V

23. W_1 is the electrostatic energy stored in a system of three equal point charges arranged in a line with 0.5 m separation between them. If W_2 is the energy stored with 1 m separation between them, then which of the following is correct?
- A) $W_1 = 0.5 W_2$
 - B) $W_1 = W_2$
 - C) $W_1 = 2 W_2$
 - D) $W_1 = 4 W_2$
24. The primary current to an ideal transformer rated at 2200/110V is 5A. The value of the turns ratio, KVA rating and the secondary current will be:
- A) 1/20, 11 kVA, and 100 A.
 - B) 1/10, 9.6 kVA, and 100 A.
 - C) 1/40, 21 kVA, and 80 A.
 - D) 1/10, 11 kVA, and 90 A
25. When a plane wave travelling in free space is incident normally on a medium having $\epsilon_r = 4.0$, the fraction of power transmitted into the medium is given by
- A) 8/9
 - B) 1/2
 - C) 1/3
 - D) 5/6
26. Which of the transformers can be used as an isolation device?
- A) Autotransformer
 - B) Ideal transformer
 - C) Linear transformer
 - D) All of the above
27. In a loaded dc generator, if the brushes are given a shift from the interpolar axis in the direction of rotation, then the commutation will:
- A) Improve with fall of terminal voltage
 - B) Deteriorate with fall of terminal voltage
 - C) Improve with rise in terminal voltage
 - D) Deteriorate with rise in terminal voltage
28. A box contains 4 white balls and 3 red balls. In succession, two balls are randomly selected and removed from the box. Given that the first removed ball is white, the probability that the second removed ball is red is:
- A) 1/3
 - B) 3/7
 - C) 1/2
 - D) 4/7

29. Given a non-uniform, positive surface charge $\rho_s = 4x^2y \text{ nC/m}^2$ on the plane $z = 4$, the total charge on the surface $-1 \leq x \leq 1$ and $-2 \leq y \leq 2$ is:
- A) Zero
 - B) $16/3 \text{ nC}$
 - C) $32/3 \text{ nC}$
 - D) $64/3 \text{ nC}$
30. The solution of the differential equation $\frac{dy}{dx} + y^2 = 0$ is
- A) $y = \frac{1}{x+c}$
 - B) $y = \frac{-x^3}{3} + c$
 - C) ce^x
 - D) Unsolvable as equation is non-linear
31. In order to match a source with internal impedance of 500Ω to a 50Ω load, the suitable transformer will be:
- A) Autotransformer
 - B) Step-up linear transformer
 - C) Step-down linear transformer
 - D) Step-down ideal transformer
32. Which of the following is the main cause of creeping in the induction type energy meters?
- A) Friction compensation
 - B) Lead/Lag Compensation
 - C) Overload Compensation
 - D) Braking torque producing system
33. A dc shunt motor finds application in
- A) Electric trains
 - B) Tape recorders
 - C) Blowers
 - D) Steel rolling mills

34. A TEM wave is incident normally upon a perfect conductor. The E and H fields at the boundary will be, respectively,
- A) Minimum and minimum
 - B) Maximum and maximum
 - C) Minimum and maximum
 - D) Maximum and minimum
35. The argument of a complex number $\frac{1+i}{1-i}$ where $i = \sqrt{-1}$, is
- A) $-\pi$
 - B) $-\pi/2$
 - C) $\pi/2$
 - D) π
36. A transmission line is short circuited at one end and has an electrical path length of $\pi/4$. If its input impedance is $j60 \Omega$, its characteristic impedance is:
- A) 60Ω
 - B) ∞
 - C) 30Ω
 - D) Data insufficient
37. The effect of stray magnetic fields on the actuating torque of a portable instrument is maximum when the operating field of the instrument and the stray fields are:
- A) Perpendicular
 - B) Parallel
 - C) Inclined at 60°
 - D) Inclined at 30°
38. A fair dice is tossed two times. The probability that the second toss results in a value that is higher than the first toss is:
- A) $2/36$
 - B) $2/6$
 - C) $5/12$
 - D) $1/2$

39. The open loop transfer function of unity feedback system is given by

$$G(s) = \frac{50}{(1 + 0.1s)(s + 10)}$$

The static error coefficients K_p , K_v and K_a respectively are:

- A) 5, 0, 0
 - B) 5, 0, ∞
 - C) 0, 0, 0
 - D) 5, ∞ , ∞
40. Torque/Weight ratio of an instrument indicates:
- A) Selectivity
 - B) Accuracy
 - C) Fidelity
 - D) Sensitivity

41. The eigen values of a skew-symmetric matrix are

- A) Always zero
- B) Always pure imaginary
- C) Either zero or pure imaginary
- D) Always real

42. A feedback system is described by the following transfer function:

$$G(s) = \frac{12}{(s^2 + 4s + 16)} \text{ and } H(s) = Ks$$

The damping factor of the system is 0.8. The overshoot of the system and the value of K respectively are:

- A) 1.3 % and 0.20
 - B) 1.5 % and 0.15
 - C) 1.3 % and 0.18
 - D) 1.5 % and 0.20
43. Breakdown torque in a 3-phase induction motor of negligible stator impedance is
- A) Directly proportional to rotor resistance
 - B) Inversely proportional to rotor resistance
 - C) Directly proportional to rotor leakage reactance
 - D) Inversely proportional to rotor leakage reactance

44. The number of turns of wire needed to provide a potentiometer with a resolution of 0.05% is:
- A) 200 turns
 - B) 2000 turns
 - C) 20 turns
 - D) 20000 turns
45. Three single-phase transformers, each of 100 kVA rating are connected in delta. If one of the transformers is taken out of service, the capacity of the system will be:
- A) 2000 Kva
 - B) 173 kVA
 - C) 115.5 kVA
 - D) 36.6 kVA
46. In reference to the Time Ratio Control Strategy of chopper, which of the following statement is false?
- A) PWM is better than FM
 - B) FM interference is an issue
 - C) Designing of filters with large frequency range in FM is easier.
 - D) Off time in FM is larger
47. The main reason for minimum air gap length in three phase induction motor is:
- A) To reduce the magnetization current of the motor.
 - B) To minimize the mutual inductance between the rotor and stator
 - C) To increase the motor physical size
 - D) None of the above
48. An 8-pole alternator runs at 750 rpm. It supplies power to a 6-pole induction motor which has a full load slip of 3%. The full load speed of the motor is:
- A) 960 rpm
 - B) 970 rpm
 - C) 1050 rpm
 - D) 1250 rpm
49. The voltage actually used for setting up the useful flux in the air-gap of a 3-phase induction motor is
- A) Equal to the Applied voltage
 - B) Greater than the Applied voltage
 - C) Less than the Applied voltage
 - D) Equal to rotor induced emf

50. In a parallel resonant circuit with $R = 100 \text{ k } \Omega$, $L = 20 \text{ mH}$, and $C = 5 \text{ nF}$, the value of resonant frequency, Quality factor and Bandwidth are:
- A) 100 krad/s, 50, and 2 krad/s
 - B) 100 krad/s, 30, and 5 krad/s
 - C) 100 krad/s, 20, and 2 krad/s
 - D) 100 krad/s, 50, and 5 krad/s

PART – II

SHORT ANSWER TYPE QUESTIONS

(Answer any 10 out of 13 questions) 10x 5 marks each = 50 marks

- 51. With a neat circuit diagram, explain the operation of a 180° variable half-wave rectifier along with its gate trigger circuit.
- 52. Calculate the depth of penetration for copper at 1 MHz. The conductivity for copper is $58 \text{ M}\Omega/\text{m}$ and the permeability is $1.26 \text{ }\mu\text{H}/\text{m}$.
- 53. Discuss the merits of delayed Automatic Gain Control (AGC) as compared with simple Automatic Gain Control (AGC).
- 54. Determine the propagation constant γ for a material having $\mu_r = 1$, $\epsilon_r = 8$ and $\sigma = 0.25 \text{ pS}/\text{m}$, if the wave frequency is 1.6 MHz
- 55. Derive a condition for maximum efficiency of a transformer.
- 56. Obtain all the possible minimal functions for: $F = \Sigma (0, 2, 3, 4, 5, 7)$.
- 57. Draw and explain the speed, torque and power-time curves for a typical electric traction drive. Briefly differentiate dynamic braking with regenerative braking
- 58. Distinguish between function generator circuits and wave shaping circuits with an example for each case.
- 59. Prove that mechanical power developed by a self-excited DC shunt motor is maximum when back emf is equal to half the applied voltage, if the field loss is neglected.
- 60. What is cut-set in a linear graph theory? Explain with an example.

61. Explain the diagram of the soft starting of induction motor to start from zero speed to rated speed. Briefly compare with a star-delta starter used for the same purpose
62. State the conditions when a 2-port network is reciprocal and when it is symmetrical. Under what condition Y-parameter of a 2-port network don not exist?
63. For measurement by ammeter and voltmeter method, what should be the accuracy of the instruments so that the overall measurement error is minimum?

PART- III

LONG ANSWER TYPE QUESTIONS

(Answer any 5 out of 8 questions) 5 x 10 marks each = 50 marks

64. What are the steps involved for the generation of gating signals for thyristors of AC to DC converters? Draw a block diagram incorporating various stages of firing circuits for a single-phase fully controlled rectifier.
65. A locker can be opened by A and B when they are both there, as the locker has 2 keys, one of which is with A and the other with B. C has both the keys, but he is permitted to open the locker only when B is not there. Show that the locker can be opened always if both A and C are there. Frame the Boolean identity that expresses this situation.
66. Show that the energy stored in a plane electromagnetic wave is equally divided between the electric and magnetic fields.
67. A single-phase is transformer working at unity power factor as an efficiency of 90% at both half full load and full load. Full load is 500 W. Determine the efficiency at 60% of full load condition.
68. Draw the circuit diagram of a half wave rectifier with a reservoir capacitor that smoothen the output. Illustrate it with output waveform. Determine the reservoir capacitor value for a HWR and smoothening circuit to supply 20 V to a load of 500 Ω . Consider maximum ripple amplitude be 10% of the average output voltage, and input signal frequency of 50 Hz
69. A fully controlled three-phase bridge is operating in inverting mode from an ac supply of 220 V, 50 Hz. The source inductance per phase is 0.32 mH. If the converter is operating with an angle of firing advance of 30° , calculate the maximum current that can be commuted, allowing a recovery angle of 5° . Neglect device volt drops.

70. A synchronous motor is running idle on an infinite bus at its rated voltage and frequency. It is excited to have an induced voltage of 125% and its synchronous impedance is $(1 + j20) \%$. Calculate the value of the stator current and its phase with respect to the bus voltage.
71. The inductive reactance of the pressure coil circuit of a dynamometer wattmeter is 0.4% of its resistance at normal frequency (50 Hz) and the capacitance is negligible. Calculate the percentage error and correction factor due to the reactance for load at 0.707 power factor lagging.

PART- IV

ESSAY TYPE QUESTIONS

(Answer any 2 out of 4 questions) 2 x 25 marks each = 50 marks

72. A 400 MVA, 22 kV star connected synchronous generator is tested for Open Circuit and Short Circuit Characteristics. The following data are obtained from these characteristics: $I_f = 1120$ A, $V_{oc} = 22$ kV, $I_{sc} = 13.2$ kA. At I_f corresponding to $I_{sc} = I_{rated}$, it is found that V_{oc} (air-gap line) = 24.2 kV (line).
- Determine X_s (Saturated) in ohms and in pu.
 - Determine SCR
 - Determine X_s (Unsaturated) and I_f corresponding to $V_{oc} = 24.2$ kV on air gap line.
73. For a unity feedback system, the open loop transfer function is given by:

$$G(s) = \frac{k}{[s(s+2)(s^2+6s+25)]}$$

- Sketch the root-locus for $0 < k < \infty$.
- At what value of k , the system becomes unstable
- At the point of instability, determine the frequency of oscillation of the system.

74. A 3-phase induction motor has an efficiency of 0.9 when the load is 37 kW. At this load, the stator copper and rotor copper loss each equals the iron loss. The mechanical losses are one-third of the no load loss. Calculate the slip.
75. Write Short Note on following:
- i. Hunting Phenomena in a Synchronous machine
 - ii. Differentiate 3-phase synchronous motor with 3-phase induction motor.
 - iii. Forward Blocking Loss and Gate power loss
 - iv. Explain how CRO is used to measure phase between two sinusoidal signals.
 - v. Buck, Boost and Buck-Boost Regulator

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SPACE FOR ROUGH WORK

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